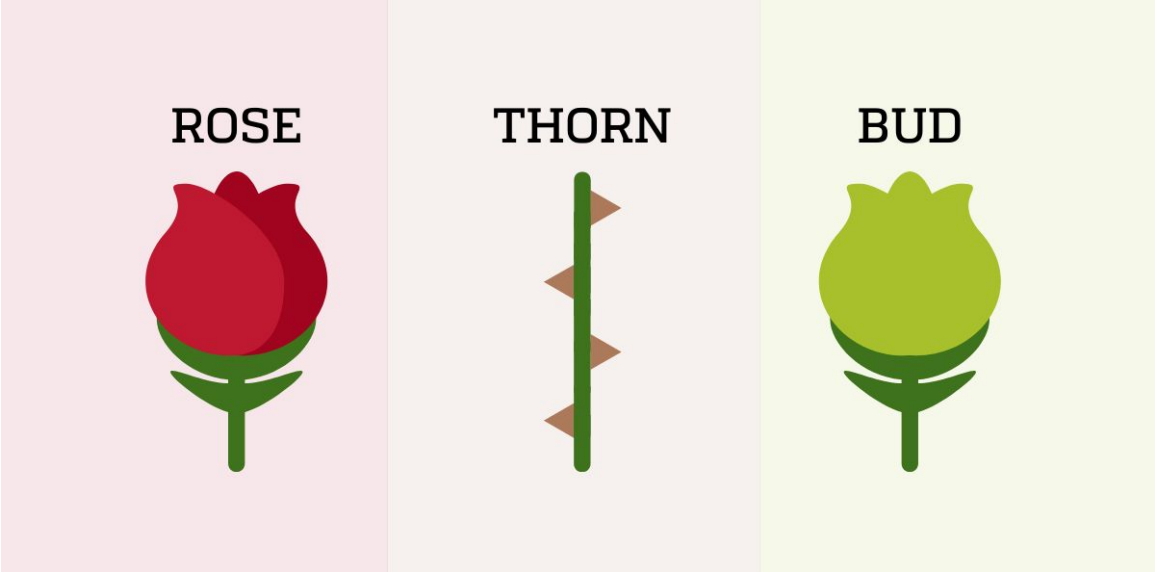
OMG I’m Doing Research Now: What to Expect

(Notes, Time Management, Questions)

2026-09-04

ROSE BUD THORN



Something positive from this week.

Something difficult from this week.

Something you're looking forward to next week.

Your grad mentors!

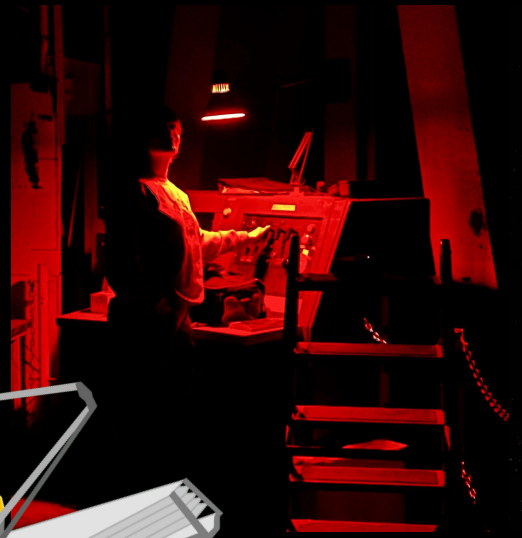
Meredith Stone

My path: KS→RI→UMass Amherst→U of A

Research: how galaxies change over time w/JWST

Around UofA: Astronomy Camp on Mt. Lemmon, TIMESTEP, SO Grad Council, AZ Science Fair

Free time: knitting and crochet, bar trivia, NHL hockey, puzzles, reading, hiking















Brian Hsu



NYC -> Taiwan -> NYC -> Boston/Harvard -> UofA

I use telescopes around the world and in space to follow up anything that explodes in the Universe!

And they trust me to operate a multi-million dollar telescope (alone)

Bok/90"!



Volleyball!



LBT!



Outside of school/work/research:

Volleyball, music festivals, Sky School, and laying in my bed

Brock Parker

Where I'm from

Grew up in Gering, NE.
Undergrad at University of Wyoming

What I research

Ultraviolet instrumentation.
Building cameras to send into space

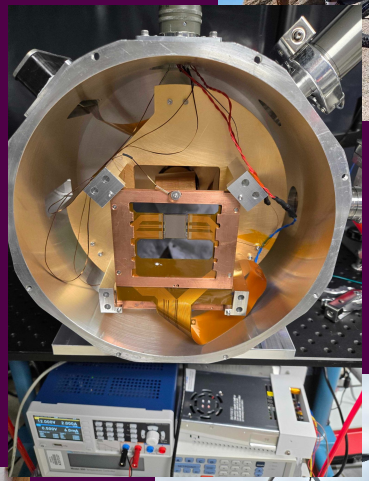
How I have fun

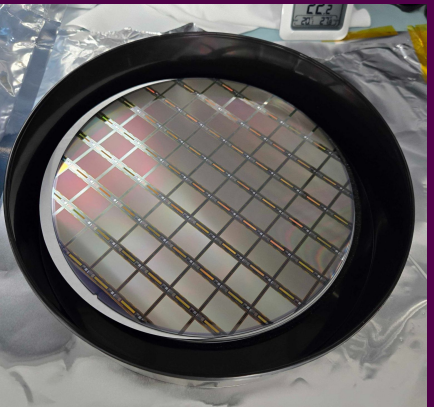
Mountain biking, hiking, climbing, baking

Pets

Rylee, Piper, Calvin













Neev Shah

What's my path been?

I'm from the **really big** city of Mumbai, on the west coast of India.

Undergrad at IISER Pune.

Spent a summer doing research in Vancouver, and visited LIGO!



What do I do?

Computational Astrophysics!

Only place where I cannot break anything (except code, which is fine)

MESA

I simulate massive stars, often in binaries to understand their lives and afterlives.

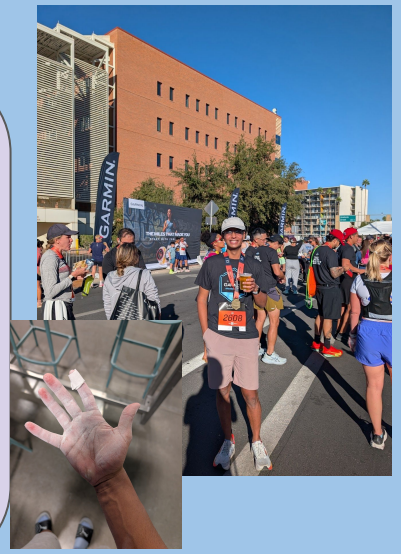
In my free time…

I love doing a lot of things poorly!

Outdoors: Running, Hiking, Bouldering (but indoors)

Indoors: watch movies at The Loft, board games, reading, eating lots of food!

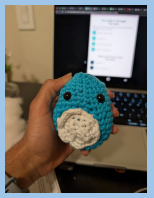
Travel: 13 countries (+1 soon)!



Pets

Do these count?











Fun fact: Although I am a theorist, I got trained to use the Bok telescope on Kitt peak (did not break!) and found my way inside all major telescopes around Tucson!

Madi VanWyngarden

Background!

I'm from Omaha, NE <3
Did my undergrad at Boston University

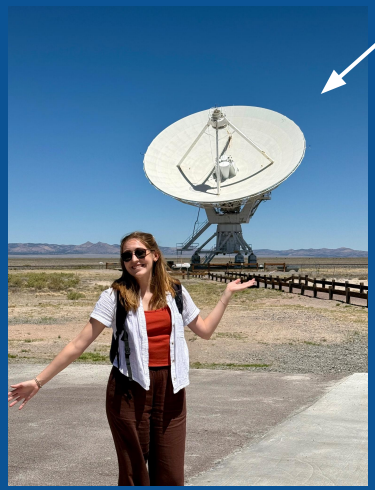




Lived in Toronto for a year doing research as a Fulbright before moving here!

Science!

I look for dusty galaxies with JWST and ALMA to study how galaxies evolve



VLA not ALMA (they won't let me go)





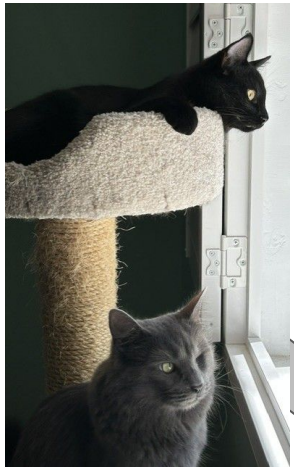


Outside of Science!

I love hiking, baking, concerts, 90s rom coms, and reading





BA, biology

GUSTAVUS
GUSTAVUS ADOLPHUS COLLEGE

Began PhD
at UA in plant pathology

Pre-K - 8th grade teacher
- Certifications in science, math, & special education
-BioMe GK-12 teacher participant

Undergraduate Research Coordinator
- TIMESTEP
- Co-I of BIORETS Santa Cruz, an NSF-funded RET with Dr. Michael Bogan



UA Science: Sky School
Assistant Director for Education
- Mountain & City programs
- Research Apprenticeship Program
- ACES Teacher PD

ARIZONA Agnese Nelms Haury Program in Environment and Social Justice

ACES
Action for Climate Equity and Sustainability

UA Science SKY SCHOOL Steward Observatory

ARIZONA WEST Water Education

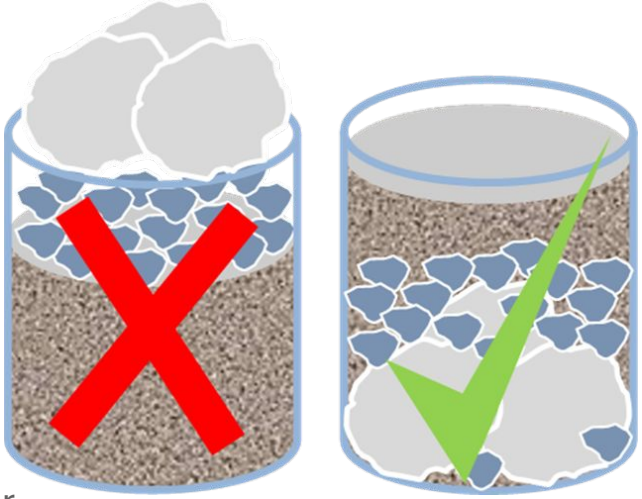
TUCSON UNIFIED SCHOOL DISTRICT

What has research looked like so far?

Any surprises?

Time Management

Pebbles and boulders



Not all tasks are equal.

- **Boulders** - writing paper, math rederivation
  I don't know how to do, far out deadlines

- **Pebbles** - Lab work, another reduction, reading another paper
  I mostly know how to do, near term deadlines

- **Sand** - sending emails, meetings, twiddling plots
  Things that happen regardless, immediate deadlines

Make time where you HAVE to work on your boulders.

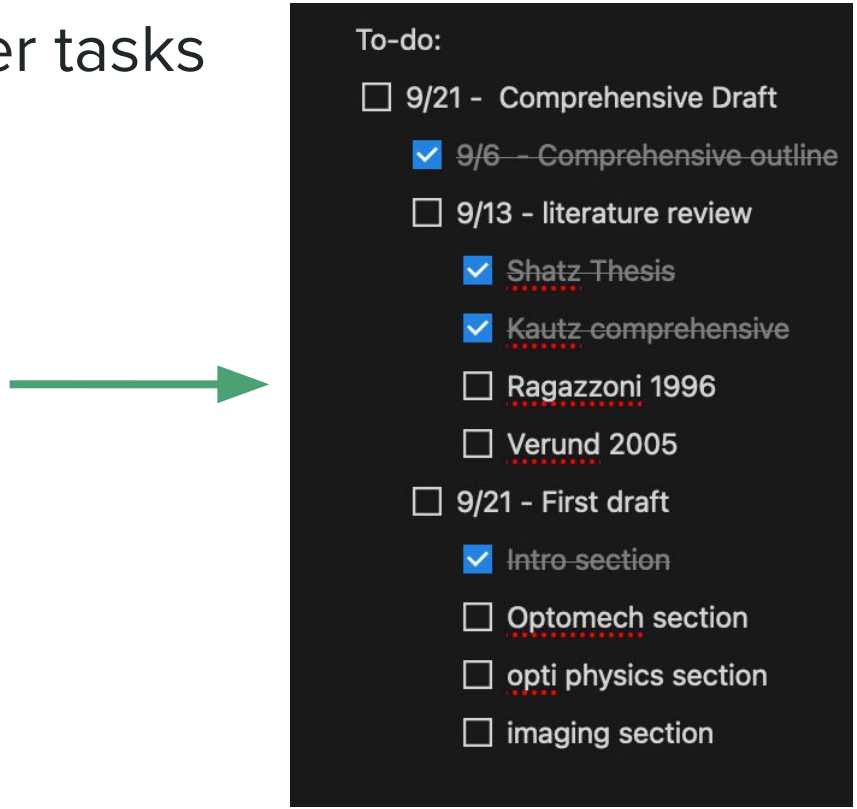
- 30 minutes of dedicated time will break the ice and make the project more manageable.
- Chip away at the larger goal
- They won't fit in your life if you let smaller tasks take up your time

Breaking down larger tasks

To-do:

- [ ] read all references
- [ ] write paper
- [ ] get PhD

I haven't completed any big tasks



But I have completed intermediate ones

Specific goals, achievable tasks (in a day or so), clear deadlines

Classes come first!

Things I had to learn the hard way

- **Code documentation.**
  Great! I wrote code for this problem 6 months ago! Oh but how does it work again...
  (spends half a day relearning own code)

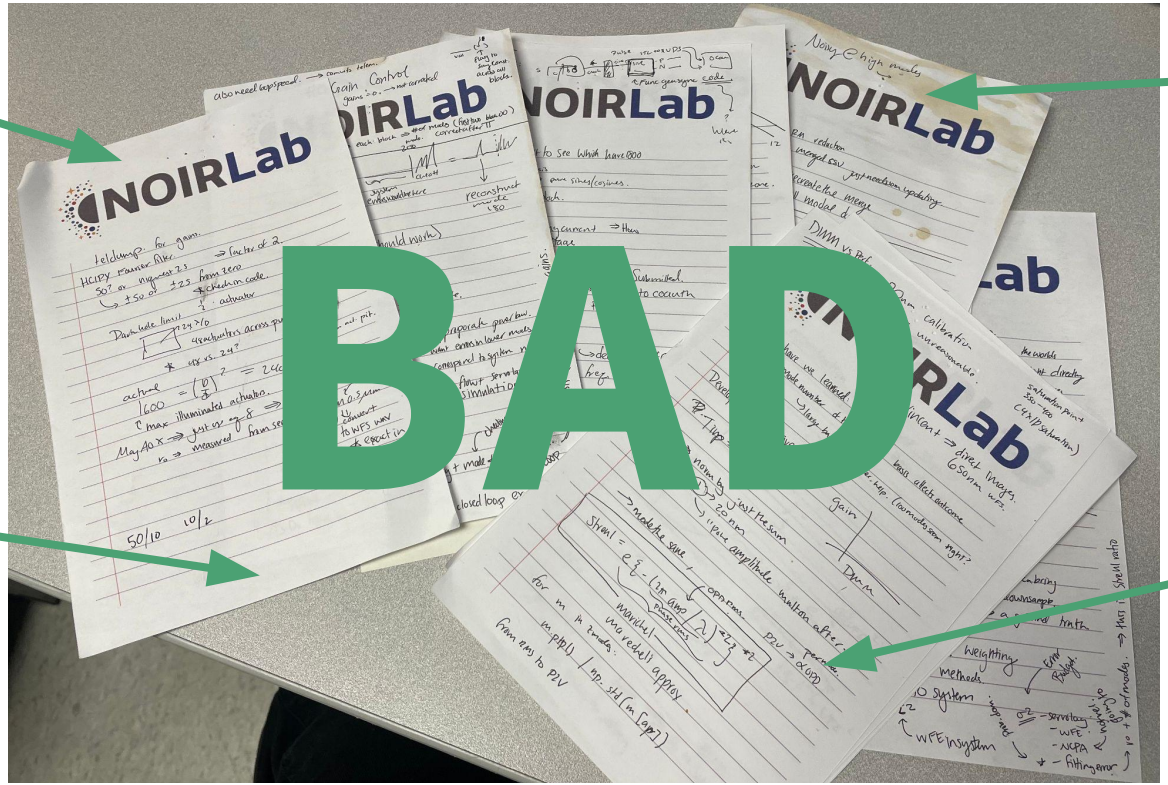
- **Bug documentation.**
  Oh I've seen this bug before! But wait, how did I fix it...
  (spends an hour on stack overflow, again)

- **Clear calibration/reduction documentation.**
  I've been using these cleaned images for months, but how on earth did I process this data...

- **Clear plot labels/documentation.**
  Supervisor asks "What are the units on the wavefront aberration in this plot?". You made the plot last year. It has no caption.

Notes Examples

Eden - random slips of paper



Meredith - python jupyter notebook

Timestamped reminders of changes, follows code progress

```
import sys
sys.path.append('~/Documents/')
import photometry_functions
```

```
datadir = os.getcwd()+'/J1120 Environment/data/'
```

```
# 2024-07-16: straight from quasar project
def get_wcs(directory):
    hdu = fits.open(directory)
    wcsi = wcs.WCS(hdu[1].header)
    hdu.close()

    return wcsi
```

```
# 2024-08-16: plotted apertures look wonky;
            # compile narrow-band and reference band (better SNR) source positions for matching
# 2024-08-19: results not great, switched reference band from F200W to F410M:
            # emphasize similar morphology to narrowband (+ SNR) over resolution
# 2024-08-19: switching back to using narrow-band for reference, F410M doesn't produce better apertures either

nb_dataref = datadir+'jw02774-o001_t002_nircam_f405n-f444w_i2d.fits'
nb_segmref = datadir+'jw02774-o001_t002_nircam_f405n-f444w_segm.fits'
mask = np.loadtxt('J1120 Environment/segm_mask.txt', dtype='bool')
nb_photref = photometry_functions.segment_phot(nb_dataref, nb_segmref, mask=None, plot_deblend=False) #maybe need to make mask

# wb_dataref = datadir+'jw02774-o001_t002_nircam_clear-f410m_i2d.fits'
# wb_segmref = datadir+'jw02774-o001_t002_nircam_clear-f410m_segm.fits'
# wb_photref = photometry_functions.segment_phot(wb_dataref, wb_segmref, mask=None, plot_deblend=False)
```

Meredith - python jupyter notebook

Markdown feature for notes, can do equations, adds to ToC on the side.

07-25-2024: trying to fix HST-JWST astrometry offset via aligning HST to JWST images: g

07-25-2024: JWST catalog looks good but resulting HST mosaic looks absolutely awful. align to Gaia/SDSS?

07-29-2024: not enough Gaia or SDSS sources to do robust matching. abandoning idea, commenting out

```
# # use f200w: high SNR and good resolution
# segment_200w = fits.getdata(os.getcwd()+'/J1120 Environment/data/jw02774-o001_t002_nircam_clear-f2
# data_shape = segment_200w.shape

# # masking locations duplicated from F405N source mask, should refine eventually b/c different pixe
# mask = np.zeros(data_shape, dtype='int')
# mask[2015:2213, 8015:8263] = 1
# mask[1008:1086, 5748:5833] = 1
# mask[2532:2613, 8086:8182] = 1
# mask[1874:2016, 1557:1745] = 1
# mask[1113:1200, 4204:4256] = 1
# mask[4285:4338, 2513:2572] = 1
# mask[0:4578, 0:100] = 1
# mask[0:4578, 4220:4360] = 1
# mask[0:4578, 5600:5747] = 1
# mask[0:4578, -100:-1] = 1
# mask[0:100, 0:10077] = 1
# mask[-100:-1, 0:10077] = 1

# np.savetxt('J1120 Environment/segm_mask_200w.txt', mask, fmt='%.1d')
```

Notes Philosophy

- Don’t just put successes, also **include failures**
  - Detail what mistakes you made to get there, how to get out of it
  - will often keep you from making the mistake again

- **Don’t assume you’ll remember** why you included something
  - Be kind to yourself, give yourself context when including plots / screenshot etc
  - Prepping a summary while you’re fully in it will help you when you explain to your mentor

- Make this a **resource** for yourself
  - Label your plots when you make them
  - Describe your plots, make them searchable (cmd-F)
  - Throw in the resources you used to make progress (links, websites, etc.)

How do you take notes?

What has worked?
What hasn't worked?
What do you want to try?

Discuss with your groups.

Research slides

Research slides?

What - a weekly slide update

Who - your research advisor 1:1, research group meetings, TIMESTEP grad meeting

Why - quickly summarize updates, communicate research process

What should a weekly research slide contain?

- Name and date
- Work accomplished that week
  - Sometimes this looks like a plot
  - Sometimes this is “reviewed literature”
  - Debugging counts
- Context for audience / future you
  - I can scroll back through my weekly updates to figure out what I was up to certain weeks
- Questions
  - Questions about work
  - Blockers - things blocking progress

This list should look like our list for notes

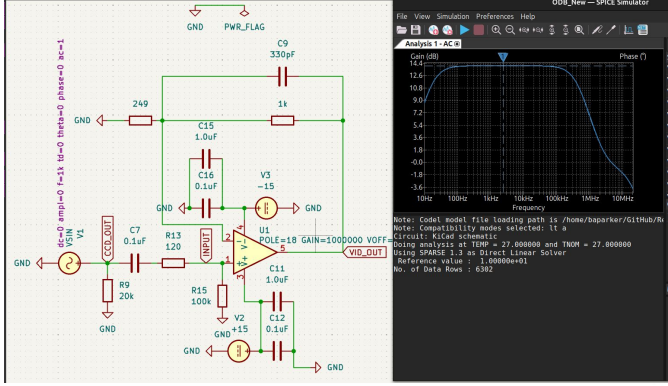
TIMESTEP
slide template

Example slides

Brock (grad)

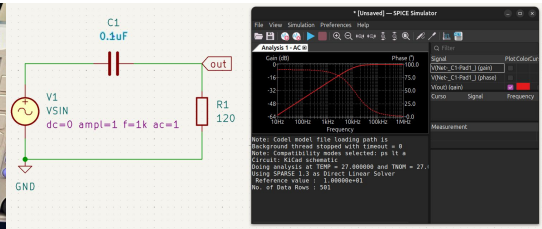
05/26/25-06/06/25

- Vacuum order
  - Changed everything to CF instead of KF, added gauge, finalized
- Lab cleaning
  - What a mess
- ODB Testing
  - Never got listed 5x amplification
  - Frequency response is the issue - cutoff frequency of 13kHz low and ~483kHz high
  - Will try to change capacitors and resistors, not sure how noise will be affected
- KiCAD
  - Simulating ODB on KiCAD, need to design new pre-amp board
  - Will make new ODB with updated parameters to have 2









06/23/25-07/04/25

- More LTA testing
  - Sequencer from Fermilab incorrect, successfully(?) repaired
  - Voltages match in sequence vs isolated, still depend on time
- Sequences correct?
  - Depends on skipper layout specifics
  - Skipping sequence good
- Full chain testing
  - No readout from skipper, even with oscilloscope
  - Problems getting data from LTA







Example slides

Neev (grad)

Week of 09/02/2026

Updates on work done last week:

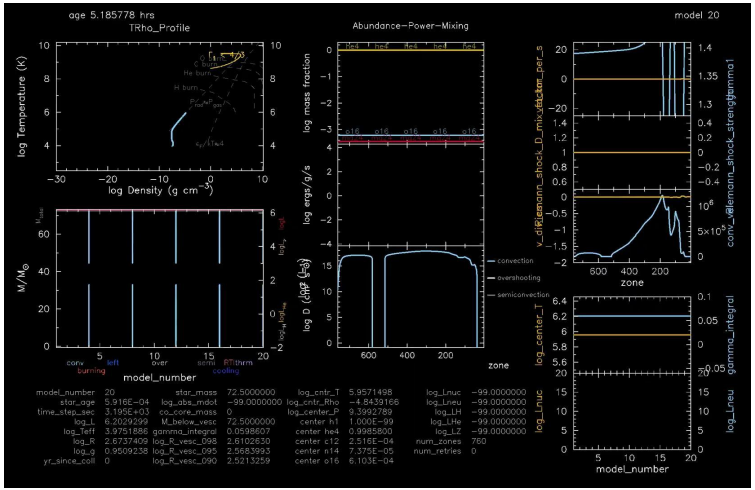
- Successfully run donor+accretor to helium depletion (with superad fix), donor + point mass (w/o superad fix)
- Attempted stripped donor/ single star in ppisn

Current roadblocks/questions:

- Accretor issues? Why superad fix worked? Eturb? Unclear what most controls are doing
- Ppisn models crash - Eb found bug in mesa
- Referee report…

Next week's steps/goals:

- Successfully run single star ppisn, donor/accretor ppisn
- Understand numerical controls and impact/use in ppisn
- Analyze output of ppisn test models / create movie



Movie of Donor starting from helium depletion - numerical issues during pulses

Example slides

Madi (grad)

Week of 08/07/26

Updates

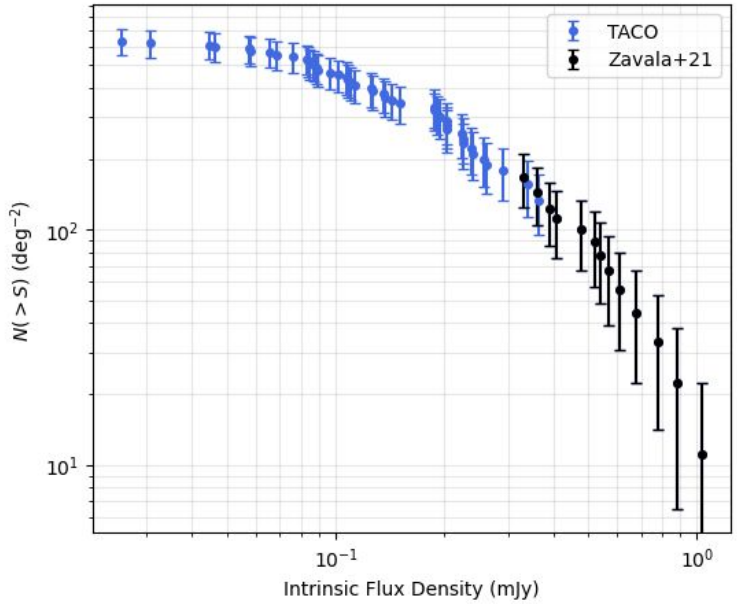
- Tried plotting the faint end extension of 2mm number counts from our survey. We are going orders of magnitude fainter than previous studies!

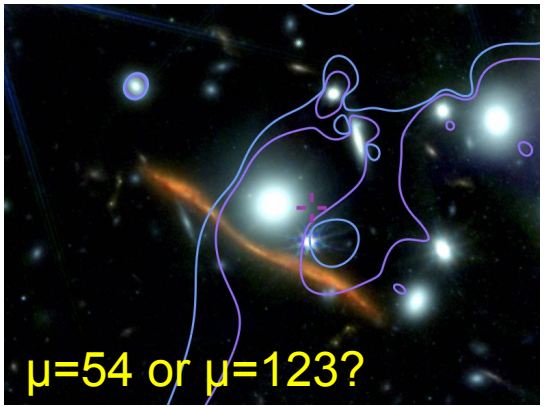
Questions

- Do I divide all points by the total combined survey area or do I treat the two surveys separately?
- There are different values in the literature for the magnification factor of the Bullet cluster sub-mm source, should I use the value from the Hershel survey paper or the VENUS lensing map?

Next Week

- Compare cumulative number counts to simulation predictions from Lagos+20

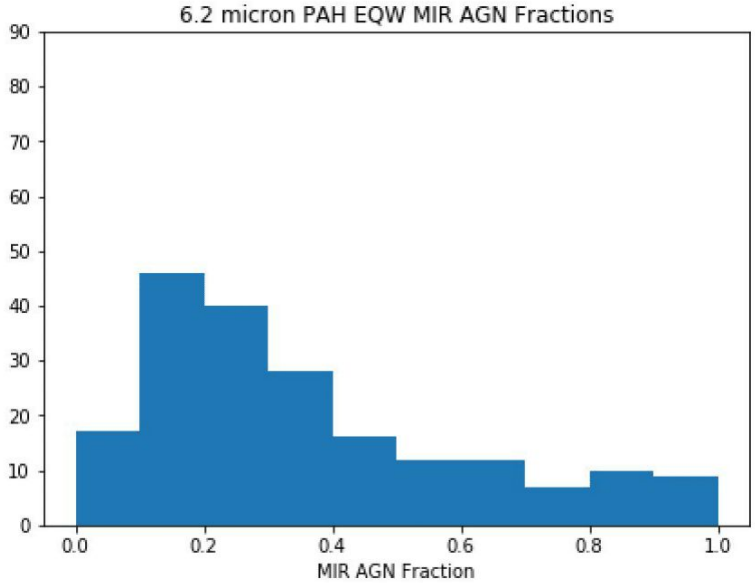


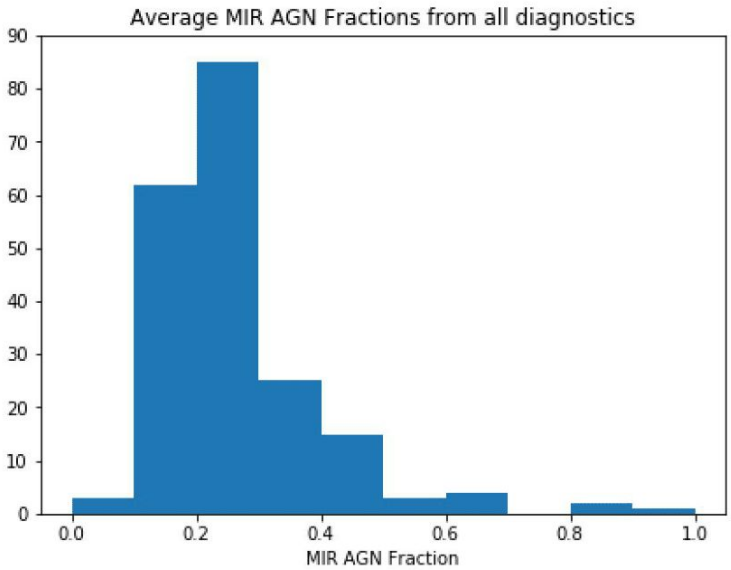


Example slides

Meredith (undergrad)

Meredith - 01/30/20





- Input new AGN fractions - very different shapes
- Working on stacking and fitting code
- Next week - determine normalization range, finish fitting code, investigate AGN lines to cut around

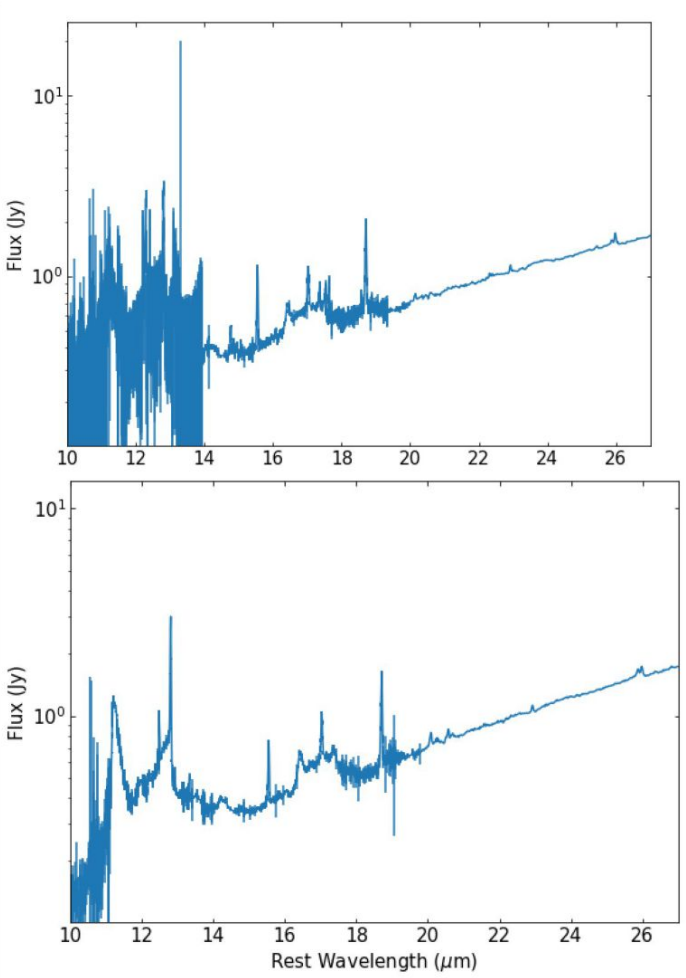
Meredith - 02/14

This week:

- Checked potential normalization window for lines - nothing
- Added sigma-clipping for the normalization (mean or median?)
- Trying to troubleshoot lowest AGN fraction stack (large errors)
- Finishing REU applications

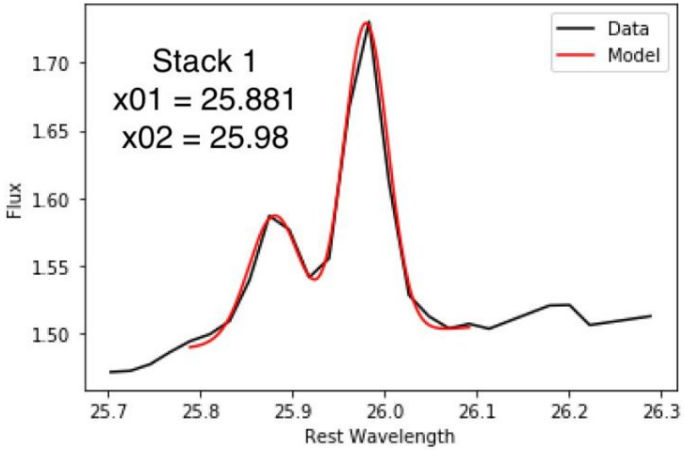
Next week:

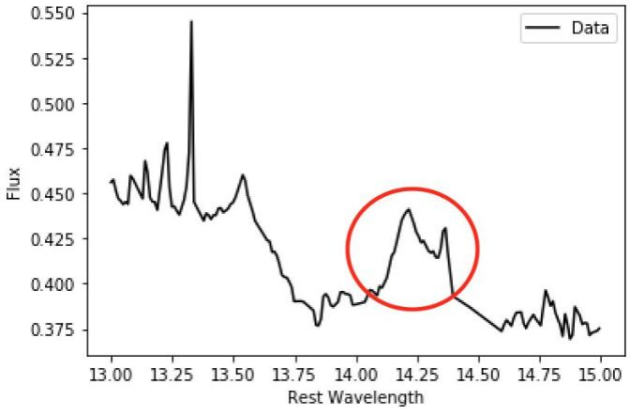
- Make non-continuum subtracted stacks
- Code continuum subtraction
- Start working on fitting



Meredith, 04/03

0.093 0.028 0.23 0.024 0.2 0.05





- "Fit" OIV/FeII, but curve_fit won't cooperate
- Not overparameterized (6 parameters, ~20 data points), right data type (64-bit), no discrete variables
- Checked individual spectra to see if NeV "bump" was present - could be seen in ~⅓, many probably high SNR
- Next week: investigate nearby AGN to see if we see the same "bump" around NeV that we see in my stacks
- Next week: calculate luminosity of NeV (and OIV if I can figure out the problem) and compare to Inami

What did you appreciate in these updates?

What did you wish they had included?

Discuss with your groups.

How to ask Questions

What is one question you currently have about your research project?

Take a minute and write / type your question down.

Anatomy of a question

~~This might be a stupid question but...~~

No question is stupid! If you're asking, it's because you're ready to learn.

Is now a good time to ask you a question?

If you’re asking a question outside of a scheduled research meeting. If the answer is “not now” ask to find a more appropriate time.

I don’t understand this paper.

You might be feeling this way, but this understanding comes slowly. Pick a few things you want to dig into.

I’ve been working through the math in **[paper name],** specifically in **[section]**. Can I review my understanding with you?

Being as specific as you can helps your mentor clue in to where you are mentally.
Asking to talk through papers is normal.

My code isn’t working.

Debugging code with a professor over your shoulder is humbling. Be specific with what help you request.

I've been working on code to [**do a thing**], but currently it [**type of error**]. So far I've tried [**debugging steps**], do you have ideas for what I could do next?

Include the steps you've already taken to work through your problem in your question. If you haven't talked in a while, remind your mentor what you've been working on

Anatomy of a Question

< Context >

< Description of problem >

< Understanding so far >

< A direct question >

Question Asking - final points

- **Getting stuck** in research is **normal**
  - Your mentors expect and want your questions!
  - It's your mentor's job to help you get unstuck

- A **well phrased question** helps both parties
  - Your mentor understands better the gaps in knowledge and request
  - You're more likely to get a helpful response

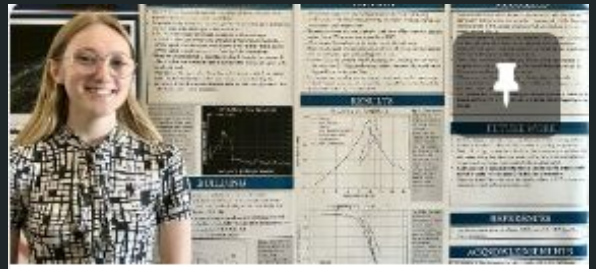
- Your **peers** are a resource
  - You're intentionally paired with another student under a mentor (where we could)
  - A TIMESTEP requirement to support each other
  - Explaining to a peer is a great test of your own knowledge

How would you rephrase your question to be more specific?

Take a minute to rewrite your question.

Then pair share with someone in your group the changes you made and why.

D2L



TIMESTEP Research Apprenticeship

2023 Non-term

Content from all Friday sessions will be saved here

Slack

Slack Etiquette

- Check every day! Mobile app helps
- Show that you've seen each comment (emojis work)
- Comment in group (apprenticeship-students-2026-27) or message people or groups of people directly

Slack 101 video (please watch!)

Physics-Astronomy Newsletter

Physics & Astronomy Newsletter

- Managed by academic advisors
- Weekly announcements (including TIMESTEP)
- Many other resources- academic plans, registration info, etc.
- Bookmark this site:

https://sites.google.com/arizona.edu/physicsastronomy/home

- Check it at least once a week for new announcements

Alumni Visit



Link also in the D2L class announcements: https://forms.gle/HcQSV7ZmvDxuHtiC6